



COMSATS University Islamabad Vehari

Department of Computer Science

Assignment 1 – Spring 2026

Subject: Programming Fundamentals (CSC-103)

Instructor: Dr. Muhammad Mudassar

Class BSCS Batch SP26

Marks=10

Course Learning Outcomes (CLOs): CLO-1 “Demonstrate the fundamental concepts of programming.”

Instructions:

1. Create an integrated PDF of programs, outputs, and answers.
2. Use CamScanner or similar software for PDF creation.
3. It is mandatory to upload the PDF to the CUOnline portal.
4. Deadline: 28 February 2026 (Saturday) till 10:00 AM.

Question 1: (Handwritten task)

- a) What is source code and machine code? How Java source code is converted to machine code? Provide details of the process.
- b) What do you understand by logical and Syntax errors? Specify errors in the following java statements.

```
System.Out.print("test");           integer price;  
float tax of product=price*0.06;
```

- c) Define and explain using suitable examples for the following term:
Keywords, Identifiers, Operators

Question 2: Arithmetic expressions. Write the pseudocode at first on the notebook and scan it to attach along with the code and then write the java code. Provide the output in the PDF file as well

- a) Write a program that reads in the radius and length of a cylinder and computes the area and volume using the following formulas:
 - $\text{area} = \text{radius} * \text{radius} * \pi$
 - $\text{volume} = \text{area} * \text{length}$.
- b) Government deducts 17% (Tax rate) as tax on the sale of mobile products, write a program that gets the package price from the user, then use the tax rate to calculate the amount of tax. Show the final amount to be paid by user to subscribe a mobile package. Also show how much tax there.
- c) Write a program that prompts the user to enter the distance to drive, the fuel efficiency of the car in miles per gallon, and the price per gallon, and displays the cost of the trip.

Question 3: Consider integer variables $x=2$, $y=5$ and $z=3$, provide the outputs

1. `System.out.println(x * y / x);`
2. `System.out.println (x * (y / x)) ;`
3. `System.out.println (y * (x / y + x / y)) ;`
4. `System.out.println (y * x / y + y * x / y);`
5. `System.out.println (3 * -2 % 4);`